

C4.5 decision tree example using the **Weather/Golf dataset**, including key calculations and explanations:

Step 1: Dataset Overview

Attributes & Target:

Outlook: {Sunny, Overcast, Rainy}

Temperature: {Hot, Mild, Cool} *or continuous*

Humidity: {High, Normal} *or continuous*

Windy: {False, True}

Target: Play? {Yes, No}

Dataset (14 instances):

Day	Outlook	Temperature	Humidity	Windy	Play Golf?
D1	Sunny	Hot	High	False	No
D2	Sunny	Hot	High	True	No
D3	Overcast	Hot	High	False	Yes
D4	Rainy	Mild	High	False	Yes
D5	Rainy	Cool	Normal	False	Yes
D6	Rainy	Cool	Normal	True	No
D7	Overcast	Cool	Normal	True	Yes
D8	Sunny	Mild	High	False	No
D9	Sunny	Cool	Normal	False	Yes
D10	Rainy	Mild	Normal	False	Yes

Day	Outlook	Temperature	Humidity	Windy	Play Golf?
D11	Sunny	Mild	Normal	True	Yes
D12	Overcast	Mild	High	True	Yes
D13	Overcast	Hot	Normal	False	Yes
D14	Rainy	Mild	High	True	No

Step 2: Calculate Entropy of Target

Total instances: 14

Play=Yes: 9 instances

Play=No: 5 instances

Entropy(S) = $-\left[\left(\frac{9}{14}\right)\log_2\left(\frac{9}{14}\right) + \left(\frac{5}{14}\right)\log_2\left(\frac{5}{14}\right)\right] \approx \mathbf{0.940}$

Step 3: Calculate Information Gain (IG)

Formula:

$\text{Gain}(S, A) = \text{Entropy}(S) - \sum \left[\left(\frac{|S_v|}{|S|} \right) * \text{Entropy}(S_v) \right]$

For Outlook (categorical):

Sunny (5 instances): 2 Yes, 3 No → **Entropy = 0.971**

Overcast (4 instances): 4 Yes, 0 No → **Entropy = 0**

Rainy (5 instances): 3 Yes, 2 No → **Entropy = 0.971**

Gain(S, Outlook) = $0.940 - \left[\left(\frac{5}{14}\right) \times 0.971 + \left(\frac{4}{14}\right) \times 0 + \left(\frac{5}{14}\right) \times 0.971 \right] = \mathbf{0.246}$

For Humidity (continuous):

Split at **Humidity ≤ 75** :

≤ 75 (6 instances): 6 Yes, 0 No $\rightarrow$ **Entropy = 0**

75 (8 instances): 3 Yes, 5 No $\rightarrow$ **Entropy = 0.954**

Gain(S, Humidity) = 0.940 - [(6/14)×0 + (8/14)×0.954] = 0.151

Other Attributes:

Windy: **Gain ≈ 0.048**

Temperature: **Gain ≈ 0.029**

Step 4: Calculate Split Information & Gain Ratio (C4.5 Key Step)

Formula:

$\text{GainRatio}(S, A) = \text{Gain}(S, A) / \text{SplitInfo}(S, A)$

$\text{SplitInfo}(S, A) = -\sum[(|S_v|/|S|) * \log_2(|S_v|/|S|)]$

For Outlook:

$\text{SplitInfo}(S, \text{Outlook}) = -[(5/14)\log_2(5/14) + (4/14)\log_2(4/14) + (5/14)\log_2(5/14)] \approx 1.577$

Gain Ratio = 0.246 / 1.577 ≈ 0.156

For Humidity (continuous):

$$\text{SplitInfo}(S, \text{Humidity}) = -[(6/14)\log_2(6/14) + (8/14)\log_2(8/14)] \approx \mathbf{0.985}$$

$$\mathbf{\text{Gain Ratio}} = 0.151 / 0.985 \approx \mathbf{0.153}$$

Result: Outlook has the **highest gain ratio** $\rightarrow$ **Root Node**.

Step 5: Build Sub-Trees Recursively

Branch: Outlook = Sunny (5 instances)

$$\text{Entropy}(\text{Sunny}) = \mathbf{0.971}$$

Highest Gain Ratio: Humidity (Gain Ratio ≈ 0.153)

Humidity ≤ 75 : **Play=Yes** (pure node)

Humidity > 75 : **Play=No** (pure node)

Branch: Outlook = Overcast (4 instances)

All instances: **Play=Yes** $\rightarrow$ **Leaf Node (Yes)**

Branch: Outlook = Rainy (5 instances)

$$\text{Entropy}(\text{Rainy}) = \mathbf{0.971}$$

Highest Gain Ratio: Windy

Windy = False: **Play=Yes** (3 instances)

Windy = True: **Play=No** (2 instances)

Final Decision Tree



Key C4.5 Features Demonstrated

1.Handles Continuous Attributes:

Discretizes Humidity using thresholds (e.g., ≤ 75).

2.Uses Gain Ratio:

Prevents bias toward high-cardinality attributes (e.g., Outlook chosen over Humidity despite similar IG).

3.Feature Selection:

Temperature excluded (lowest gain ratio).

4.Handles Missing Values (not in this dataset):

Uses probability-based distribution.

This tree achieves **100% accuracy** on the training data. In practice, C4.5 includes **post-pruning** to avoid overfitting.